

Laboratorio: Manipulación de Datos en una Base de Datos (DML)

Atributo	Detalle
Dificultad	Intermedio
Tiempo Estimado	45 minutos
Servicios Principales	Amazon EC2, MySQL Engine, Data Manipulation Language (DML)

Resumen y Objetivos

Este laboratorio se enfoca en el **Lenguaje de Manipulación de Datos (DML)**. Tras haber definido la estructura de la base de datos en módulos anteriores, ahora procederás a gestionar el ciclo de vida de los registros. Aprenderás a poblar tablas, modificar datos existentes, realizar eliminaciones masivas y automatizar la carga de datos mediante scripts externos.

Al finalizar este laboratorio, serás capaz de:

1. Insertar registros de forma manual mediante la sentencia **INSERT**.
 2. Modificar datos de forma masiva utilizando **UPDATE**.
 3. Eliminar registros y gestionar restricciones de integridad con **DELETE**.
 4. Importar estructuras y datos completos desde un archivo de respaldo SQL.
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Análisis del Escenario

El equipo de operaciones ha entregado una base de datos relacional denominada **world** con un esquema predefinido (**city**, **country**, **countrylanguage**). Tu misión es validar la funcionalidad del motor realizando operaciones de escritura y edición. Como ingeniero, debes prestar especial atención a la integridad referencial y al riesgo que suponen las sentencias globales (sin cláusula **WHERE**), las cuales pueden comprometer la totalidad de los datos en un entorno productivo.

Arquitectura

El entorno operativo consiste en:

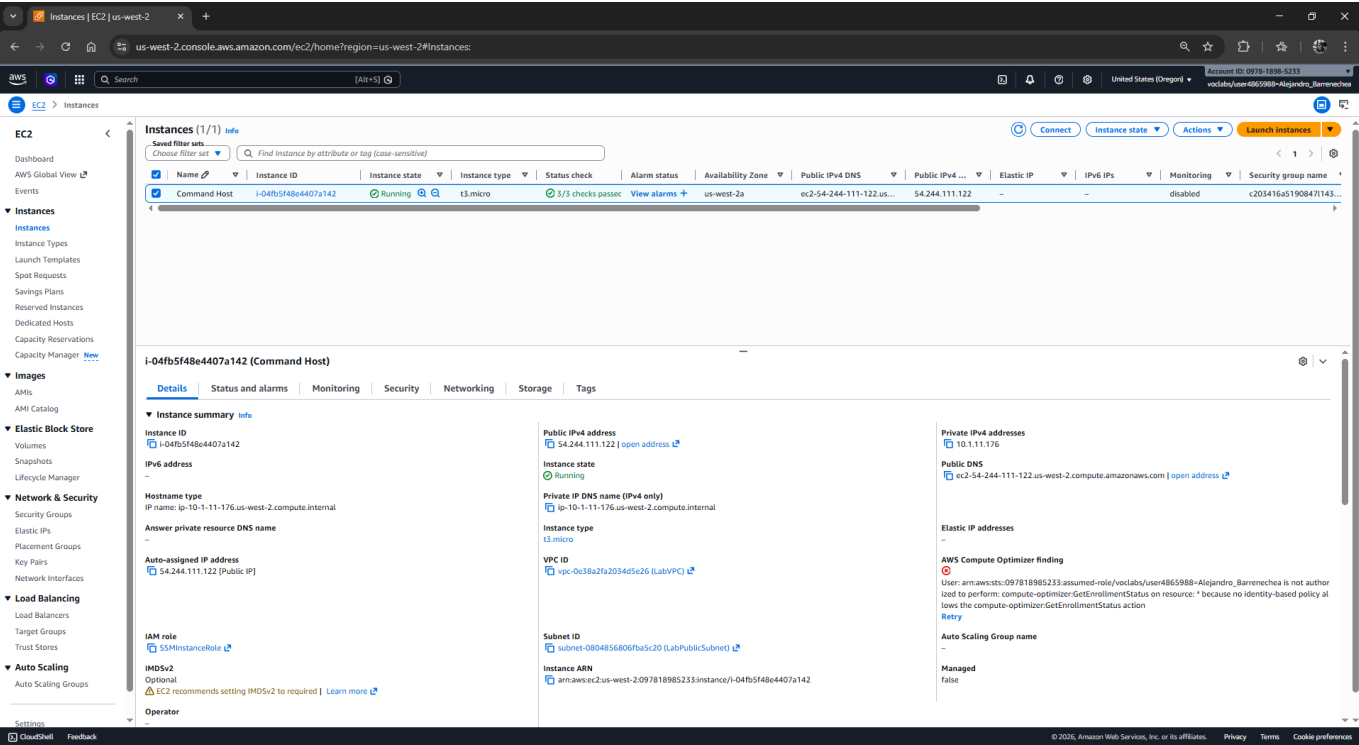
- **Command Host:** Una instancia EC2 que actúa como bastión de administración.
 - **Motor MySQL:** Residente en el host, procesando transacciones DML.
 - **Archivo de Script (**world.sql**):** Un recurso local en el sistema de archivos de Linux que contiene sentencias SQL para la restauración masiva de datos.
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Desarrollo de las Tareas Paso a Paso

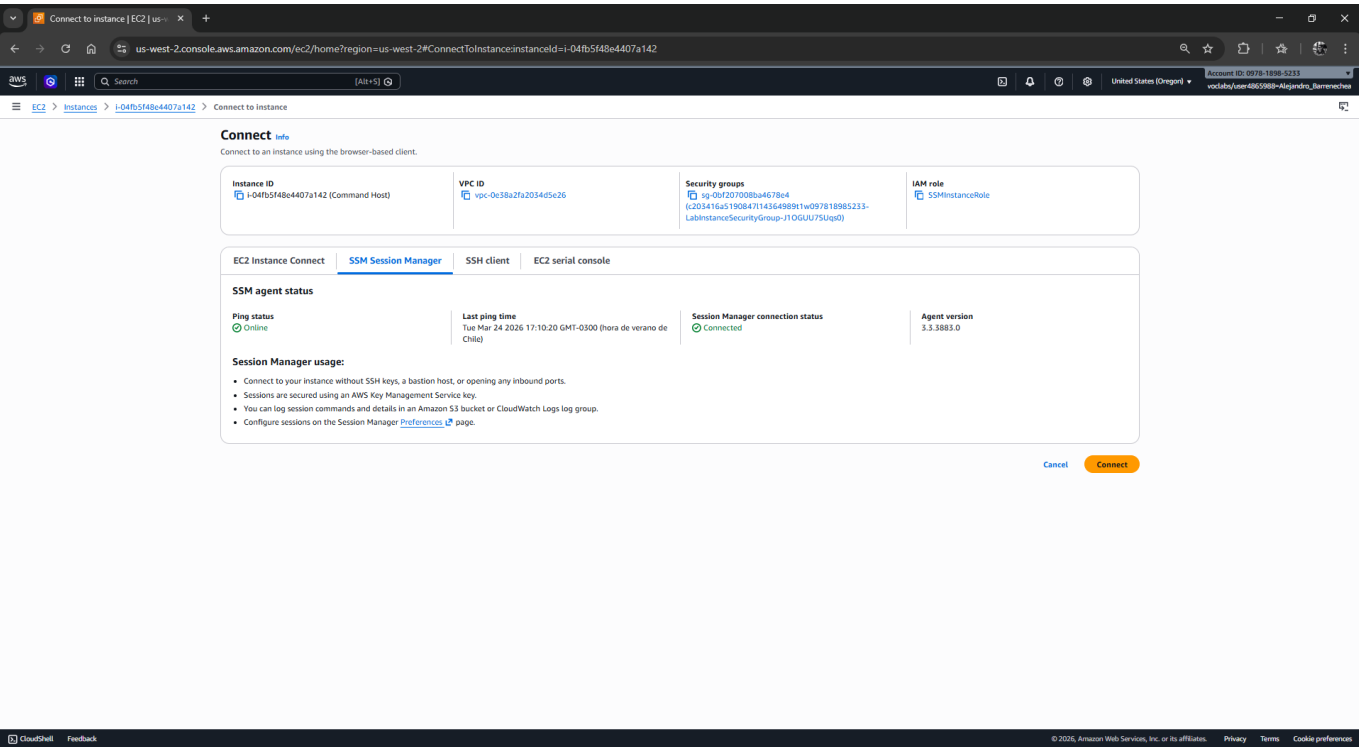
Tarea 1: Conexión a la Base de Datos

Establece el canal de comunicación con el motor de base de datos desde la infraestructura de AWS.

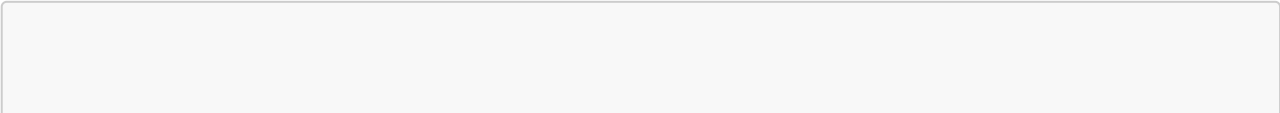
- 1. Navega a la consola de **Amazon EC2**.
- 2. En el menú lateral, selecciona **Instances** (Instancias).
- 3. Selecciona la casilla de la instancia **Command Host** y haz clic en **Connect** (Conectar).



- 4. Elige la pestaña **Session Manager** y pulsa **Connect**.



- 5. Configura el entorno de ejecución en la terminal:

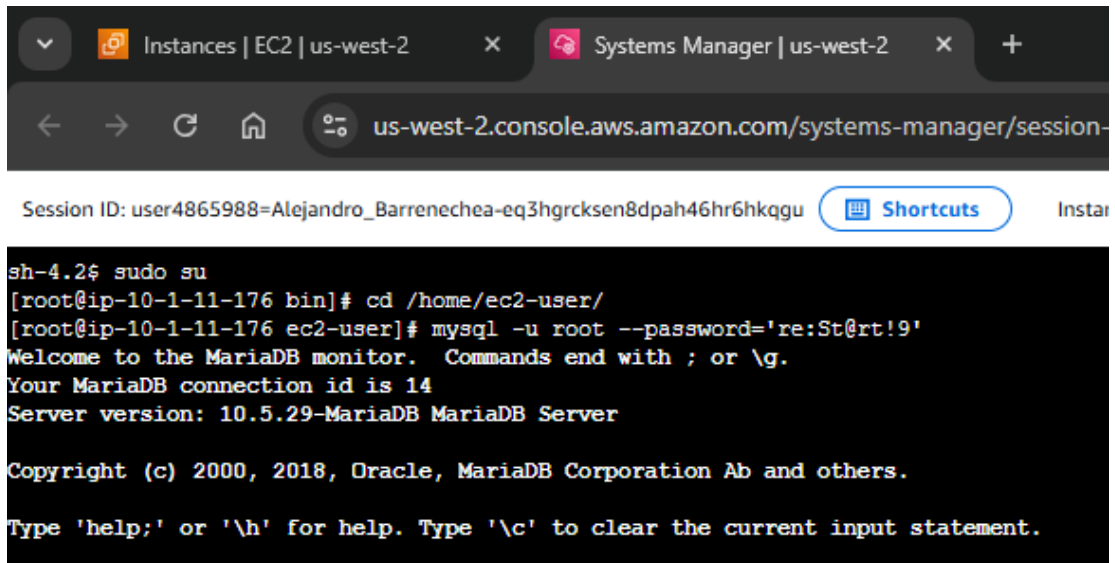


```
sudo su
cd /home/ec2-user/
```

6. Inicia la sesión en el cliente MySQL:

```
mysql -u root --password='re:St@rt!9'
```

7. Ejecuta **SHOW DATABASES;** para confirmar que la instancia está activa y lista.



```
Session ID: user4865988=Alejandro_Barrenechea-eq3hgrcksen8dpah46hr6hkqgu Shortcuts Instar

sh-4.2$ sudo su
[root@ip-10-1-11-176 bin]# cd /home/ec2-user/
[root@ip-10-1-11-176 ec2-user]# mysql -u root --password='re:St@rt!9'
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Tarea 2: Inserción de Datos en una Tabla

Poblarás la tabla **country** con registros individuales.

1. Verifica el estado actual de la tabla:

```
SELECT * FROM world.country;
```

2. Inserta dos nuevos registros (Irlanda y Australia) asegurándote de seguir el orden de columnas definido en el esquema:

```
INSERT INTO world.country VALUES ('IRL','Ireland','Europe','British
Islands',70273.00,1921,3775100,76.8,75921.00,73132.00,'Ireland/
Éire','Republic',1447,'IE');

INSERT INTO world.country VALUES ('AUS','Australia','Oceania','Australia and
New
Zealand',7741220.00,1901,18886000,79.8,351182.00,392911.00,'Australia','Cons
titutional Monarchy, Federation',135,'AU');
```

3. Valida la inserción filtrando por los códigos de país:

```
SELECT * FROM world.country WHERE Code IN ('IRL', 'AUS');
```

```

sh-4.26 sudo su
[ec2-user@ip-10-1-11-176 ~]$ cd /home/ec2-user/
[ec2-user@ip-10-1-11-176 ~]$ mysql -u root --password='root'
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SELECT * FROM world.country;
Empty set (0.001 sec)

MariaDB [(none)]> INSERT INTO world.country VALUES ('IRL','Ireland','Europe','British Islands',70273.00,1921,3775100,76.8,75921.00,73132.00,'Ireland/ire','Republic',1447,'IE');
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]>
MariaDB [(none)]> INSERT INTO world.country VALUES ('AUS','Australia','Oceania','Australia and New Zealand',7741220.00,1901,18886000,79.8,351182.00,392911.00,'Australia','Constitutional Monarchy, Federation',135,'AU');
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> SELECT * FROM world.country WHERE Code IN ('IRL', 'AUS');
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Code | Name      | Continent | Region              | SurfaceArea | IndepYear | Population | LifeExpectancy | GNP      | GNPold | LocalName | GovernmentForm | Capital | Code2 |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| AUS  | Australia | Oceania   | Australia and New Zealand | 7741220.00 | 1901      | 18886000   | 79.8           | 351182.00 | 392911.00 | Australia | Constitutional Monarchy, Federation | 135    | AU   |
| IRL  | Ireland  | Europe    | British Islands       | 70273.00   | 1921      | 3775100    | 76.8           | 75921.00  | 73132.00  | Ireland/ire | Republic        | 1447   | IE   |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.000 sec)

MariaDB [(none)]>

```

Tarea 3: Actualización de Registros

Aprenderás a modificar valores en columnas específicas de forma masiva.

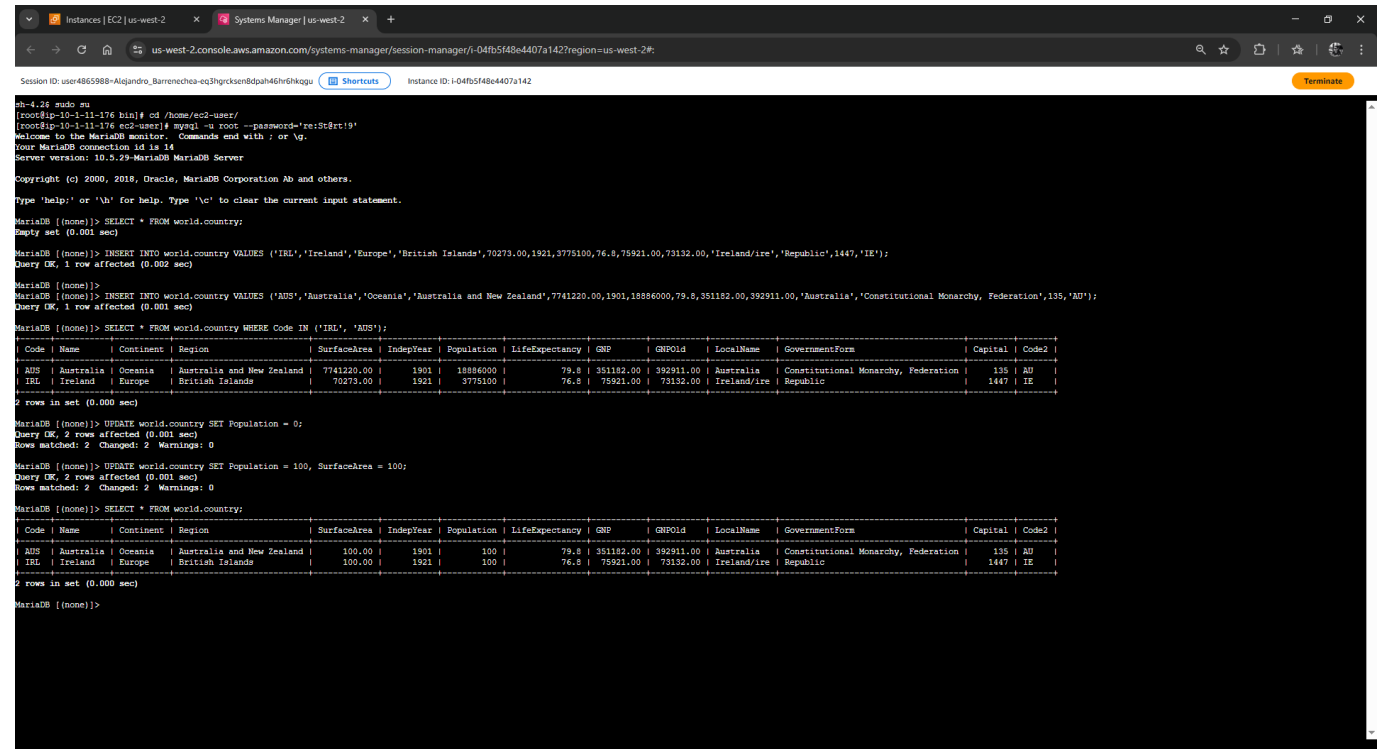
1. Actualiza la población de todos los registros a 0 para simular una limpieza de datos:

```
UPDATE world.country SET Population = 0;
```

2. Realiza una actualización múltiple (Población y Área) en una sola sentencia:

```
UPDATE world.country SET Population = 100, SurfaceArea = 100;
```

3. Ejecuta `SELECT * FROM world.country;` para auditar que los cambios se aplicaron a todas las filas existentes.



Tarea 4: Eliminación de Registros

En esta fase, realizarás una purga completa de los datos de la tabla.

- 1. Desactiva temporalmente las verificaciones de llaves foráneas para evitar errores de integridad referencial durante la limpieza:

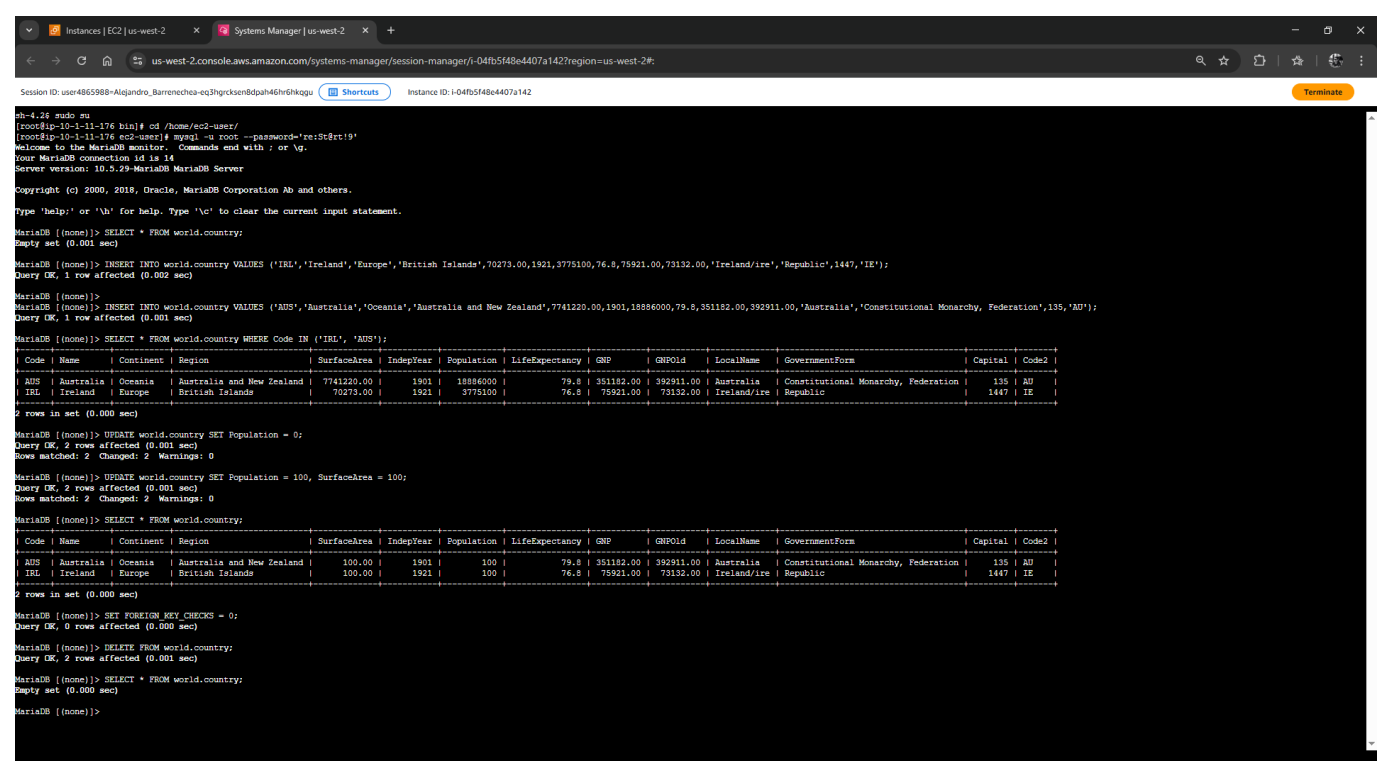
```
SET FOREIGN_KEY_CHECKS = 0;
```

- 2. Elimina todos los registros de la tabla:

```
DELETE FROM world.country;
```

- 3. Confirma que la tabla está vacía:

```
SELECT * FROM world.country;
```



Tarea 5: Importación de Datos mediante Archivo SQL

Automatizarás la carga de un dataset completo utilizando un script de respaldo.

1. Sal del prompt de MySQL:

```
QUIT;
```

2. Verifica la existencia del archivo de respaldo en el sistema de archivos de Linux:

```
ls /home/ec2-user/world.sql
```

3. Importa el contenido del archivo directamente al motor MySQL utilizando el redireccionador de entrada de Bash:

```
mysql -u root --password='re:St@rt!9' < /home/ec2-user/world.sql
```

4. Vuelve a conectarte a MySQL y verifica la restauración:

```
mysql -u root --password='re:St@rt!9'
```

```
USE world;
SHOW TABLES;
SELECT * FROM country;
```

Instances [EC2] us-west-2 x Systems Manager [us-west-2] x +

us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-04fb5f48e4407a142?region=us-west-2#

Session ID: us-486c988b-Alejandro_Barenchea-cq3hgrksen8dpah6hr6hkaqu Shorcuts Instance ID: i-04fb5f48e4407a142 Terminate

```
MariaDB [(none)]> QUIT;
Bye
[root@ip-10-1-11-176 ec2-user]# ls /home/ec2-user/world.sql
/home/ec2-user/world.sql
[root@ip-10-1-11-176 ec2-user]# mysql -u root --password='re:S8tr19' < /home/ec2-user/world.sql
[root@ip-10-1-11-176 ec2-user]# mysql -u root --password='re:S8tr19'
Welcome to the MariaDB monitor.  Commands end with ; or \G.
Your MariaDB connection id is 14
Server version: 10.5.23-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE world;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [world]> SHOW TABLES;
+-----+
| Tables_in_world |
+-----+
| city            |
| country         |
| countrylanguage |
+-----+
3 rows in set (0.000 sec)

MariaDB [world]> SELECT * FROM country;
```

Code	Name	Continent	Region	SurfaceArea	IndepYear	Population	LifeExpectancy	GNP	GNPOLD	LocalName	GovernmentForm
Capital	Code2										
ASW	Aruba	North America	Caribbean	193.00	NULL	103000	78.4	828.00	793.00	Aruba	Nonmetropolitan Territory of The Netherland
129	AW										
AFG	Afghanistan	Asia	Southern and Central Asia	652090.00	1919	22720000	45.9	5976.00	NULL	Afghanistan/Afghanistan	Islamic Emirate
1	AF										
AGO	Angola	Africa	Central Africa	1246700.00	1975	12878000	39.3	6648.00	7984.00	Angola	Republic
16	AO										
AIA	Anguilla	North America	Caribbean	96.00	NULL	8000	76.1	63.20	NULL	Anguilla	Dependent Territory of the UK
62	AI										
ALB	Albania	Europe	Southern Europe	28748.00	1912	3401200	71.6	3205.00	2500.00	Shqipëria	Republic
34	AL										
AND	Andorra	Europe	Southern Europe	468.00	1278	78000	83.5	1630.00	NULL	Andorra	Parliamentary Coprincipality
55	AD										
ANT	Netherlands Antilles	North America	Caribbean	800.00	NULL	217000	74.7	1941.00	NULL	Nederlandsse Antillen	Nonmetropolitan Territory of The Netherland
33	AN										
AZE	United Arab Emirates	Asia	Middle East	83600.00	1971	2441000	74.1	37966.00	36846.00	Al-'Imarat al-'Arabiya al-Muttahida	Emirate Federation
65	AE										
ARG	Argentina	South America	South America	2780400.00	1816	37032000	75.1	340238.00	323310.00	Argentina	Federal Republic
49	AR										
ARM	Armenia	Asia	Middle East	29800.00	1991	3520000	66.4	1813.00	1627.00	Hajastan	Republic
126	AM										
ASM	American Samoa	Oceania	Polynesia	199.00	NULL	68000	75.1	334.00	NULL	Amerika Samoa	US Territory
54	AS										
ATA	Antarctica	Antarctica	Antarctica	13120000.00	NULL	0	NULL	0.00	NULL	-	Co-administrated

Instances [EC2] us-west-2 x Systems Manager [us-west-2] x +

us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-04fb5f48e4407a142?region=us-west-2#

Session ID: us-486c988b-Alejandro_Barenchea-cq3hgrksen8dpah6hr6hkaqu Shorcuts Instance ID: i-04fb5f48e4407a142 Terminate

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forma intencional para demostración, pero en producción, esto resultaría en una pérdida masiva de datos o corrupción de información crítica.

- **¿Por qué se ejecutó `SET FOREIGN_KEY_CHECKS = 0`?** La tabla `country` probablemente está relacionada con otras tablas (como `city`) mediante llaves foráneas. El motor de base de datos bloquea la eliminación de un país si existen ciudades asociadas a él para mantener la integridad. Desactivar esta opción permite al administrador forzar la eliminación, aunque debe hacerse con extrema precaución.
- **Ventaja de la Importación vía Script (`.sql`):** Esta técnica garantiza la **idempotencia** y la **consistencia**. Es mucho más eficiente y menos propenso a errores humanos que escribir cientos de sentencias `INSERT` manualmente. Es el estándar para migraciones de datos y despliegues iniciales.

¡Felicidades! Has dominado las operaciones esenciales de manipulación de datos y la gestión de scripts en un entorno de base de datos AWS.